

Practice Exercise Encapsulation Plus

Design and Implement a Java program as follows:

1) Create CoveredInbox class as follows

- Attributes to keep track when inbox is opened/closed and how many letters it has
- Functionality:
 - Get number of letters - fails if inbox is still closed
 - Open inbox - fails if already opened
 - Close inbox - fails if already closed
 - Add letter to inbox (increments number of letters) - fails if inbox is closed
 - Remove letter from inbox (decrements number of letters) - fails if inbox is closed

2) Create RunInbox class with main method which stores the instance of inbox

3) In main method display a menu where the user can:

- 1. Open inbox
- 2: Close inbox
- 3: Add letter to inbox
- 4: Remove letter from inbox
- 5: Check number of letters in inbox
- 9: Exit program - fail if inbox is still open

4) Make sure to use encapsulation - private attributes, get methods, and appropriate set methods

SAMPLE RUN 1

```
MENU
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program

Enter your selection : 4

Cannot remove letter - inbox is empty
```

SAMPLE RUN 2

```
MENU
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program

Enter your selection : 5

Inbox is closed - cannot check number of letters
```

SAMPLE RUN 3

```
MENU
```

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 3

Cannot add letter - inbox is still closed

SAMPLE RUN 4

MENU

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 1

MENU

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 3

MENU

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 5

Letters in inbox =1

SAMPLE RUN 5

MENU

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 1

MENU

```
1: Open inbox
2: Close inbox
3: Add letter to inbox
4: Remove letter from inbox
5: Check number of letters in inbox
9: Exit program
```

Enter your selection : 9

You need to close inbox before exiting

MENU

- 1: Open inbox
- 2: Close inbox
- 3: Add letter to inbox
- 4: Remove letter from inbox
- 5: Check number of letters in inbox
- 9: Exit program

Enter your selection : 2

MENU

- 1: Open inbox
- 2: Close inbox
- 3: Add letter to inbox
- 4: Remove letter from inbox
- 5: Check number of letters in inbox
- 9: Exit program

Enter your selection : 9

Thank you for using the program. Goodbye!

Sample solution

```
public class CoveredInbox {

    // attributes are private
    private boolean open;
    private int numLetters;

    // get methods provide interface and do validation as needed
    public boolean isOpen() {
        return open;
    }
    public int getNumLetters() {
        if (!open)
            System.out.println("\nInbox is closed - cannot check number of letters");
        return numLetters;
    }

    // no set methods because functional methods set values

    // method to open cover on the inbox
    public void openCover() {
        if (open)
            System.out.println("\nInbox is already opened");
        else
            open = true;
    }

    // method to close the cover on the inbox
    public void closeCover() {
        if (!open)
            System.out.println("\nInbox is already closed");
        else
            open = false;
    }

    // method to add a letter to the inbox
    public void addLetter() {
        if (!open)
            System.out.println("\nCannot add letter - inbox is still closed");
        else
            numLetters++;
    }

    // method to remove letter from inbox
    public void removeLetter() {
        if (numLetters == 0)
            System.out.println("\nCannot remove letter - inbox is empty");
        else if (!open)
            System.out.println("\nCannot remove letter - inbox is still closed");
        else
            numLetters--;
    }
}

import java.util.Scanner;

public class RunInbox {

    private CoveredInbox inbox; // instance of the inbox
```

```

// Constructor
public RunInbox() {
    inbox = new CoveredInbox();
}

// method to display the menu
public void displayMenu() {

    System.out.println("\n    MENU");
    System.out.println("1: Open inbox ");
    System.out.println("2: Close inbox ");
    System.out.println("3: Add letter to inbox ");
    System.out.println("4: Remove letter from inbox ");
    System.out.println("5: Check number of letters in inbox ");
    System.out.println("9: Exit program");

}

// method to handle user's selection
public void processChoice(int c) {

    switch (c) {

        case 1 :      inbox.openCover();
                     break;
        case 2 :      inbox.closeCover();
                     break;
        case 3 :      inbox.addLetter();
                     break;
        case 4 :      inbox.removeLetter();
                     break;
        case 5 :      System.out.println("\nLetters in inbox = " +
inbox.getNumLetters());
                     break;
        case 9:      if (inbox.isOpen())
                        System.out.println("\nYou need to close inbox
before exiting");
                     else
                        System.out.println("\nThank you for using the
program. Goodbye!");
                     break;
        default:      System.out.println("Invalid choice");

    }

}

public static void main(String[] args) {

    RunInbox run = new RunInbox(); // new instance

    Scanner stdin = new Scanner(System.in);
    int selection = 0;

    do {

        run.displayMenu();

        System.out.print("\nEnter your selection : ");
        selection = stdin.nextInt();

        run.processChoice(selection);

    } while (selection != 9 || (selection == 9 && run.inbox.isOpen() == true));

    stdin.close();

}
}

```